FLATNESS OF THE LIFTED MORPHISM IN A MORPHISM OF THICKENINGS OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4023-4041 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 4043-4064, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Morphisms of thickenings, flatness over a base, and etale charts for algebraic spaces; the reduction and descent are retained.

Source reference. Original lemma more-morphisms-lemma-flatness-morphism-thickenings: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4064–4064. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Consider a commutative diagram*

$$\begin{array}{ccc} (X \subset X') & \xrightarrow[(f,f')]{} & (Y \subset Y') \\ & \searrow \quad \swarrow & \\ & (B \subset B') & \end{array}$$

of thickenings of algebraic spaces over S. Assume

(1) *X' is flat over B',*
(2) *f is flat,*
(3) *$B \subset B'$ is a finite order thickening, and*
(4) *$X = B \times_{B'} X'$ and $Y = B \times_{B'} Y'$.*

Then f' is flat and Y' is flat over B' at all points in the image of f'.

Proof. Choose a scheme U' and a surjective étale morphism $U' \to B'$. Choose a scheme V' and a surjective étale morphism $V' \to U' \times_{B'} Y'$. Choose a scheme W' and a surjective étale morphism $W' \to V' \times_{Y'} X'$. Let U, V, W be the base change of U', V', W' by $B \to B'$. Then flatness of f' is equivalent to flatness of $W' \to V'$

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and we are given that $W \to V$ is flat. Hence we may apply the lemma in the case of schemes to the diagram

$$\begin{array}{ccc} (W \subset W') & \longrightarrow & (V \subset V') \\ & \searrow \quad \swarrow & \\ & (U \subset U') & \end{array}$$

of thickenings of schemes. See More on Morphisms, Lemma more-morphisms-lemma-flatness-morphism-thickenings. The statement about flatness of Y'/B' at points in the image of f' follows in the same manner. $\square$